



Module 11: Database Design

CPE 010 Database Management System

Slides based on “Database Systems: Design, Implementation, and Management 8th Edition” Chapter 9:
Database Design Slides by Rob, P. and Coronel, C.

Intended Learning Outcomes

After this module, the student should be able to:

- Design databases using the phases prescribed in the SDLC and DBLC
- Modify SDLC and DBLC frameworks for revisions
- Create and execute stored procedures using Management Studio and T-SQL

The Information System

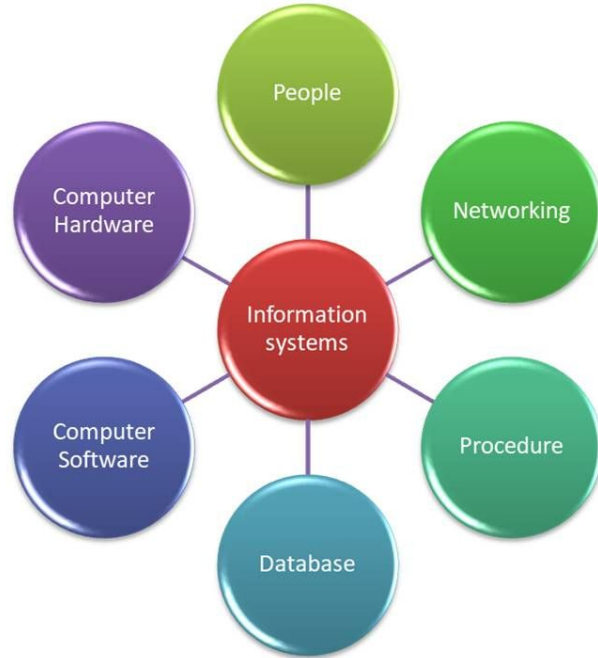
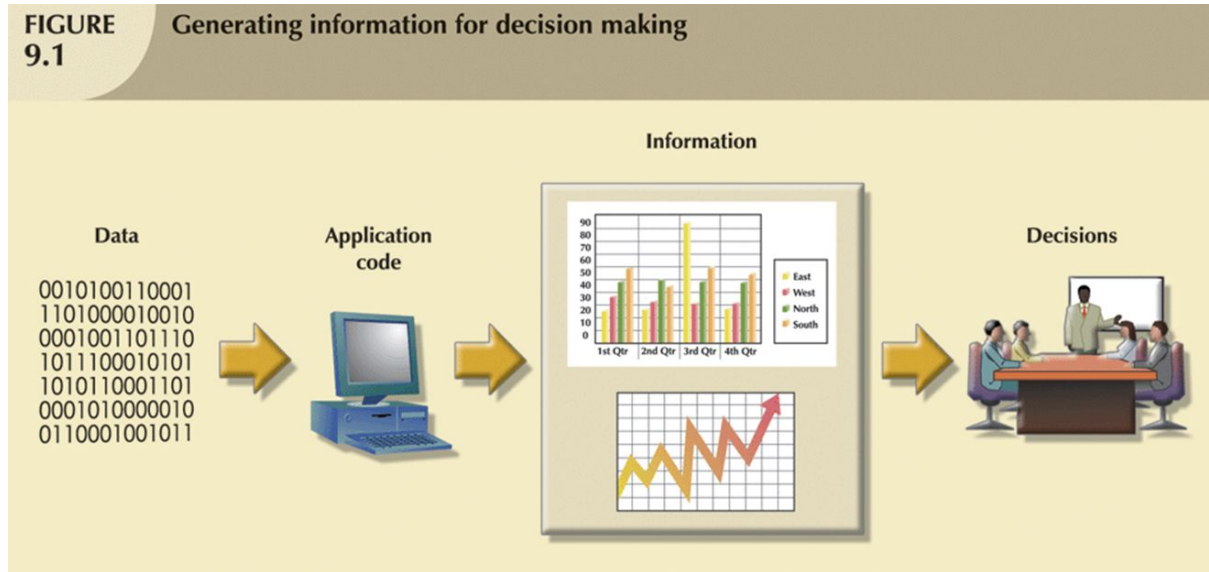


Image Source: <https://www.knowcomputing.com/>

- Composed of:
 - People, hardware, software
 - Database(s), application programs, procedures
- Systems analysis
 - Process that establishes need for and extent of information system
- Systems development
 - Process of creating information system

Applications: Transform data into information that forms basis for decision making.

FIGURE 9.1 Generating information for decision making



Composed of following two parts:

- Data
- Code by which data are transformed into information

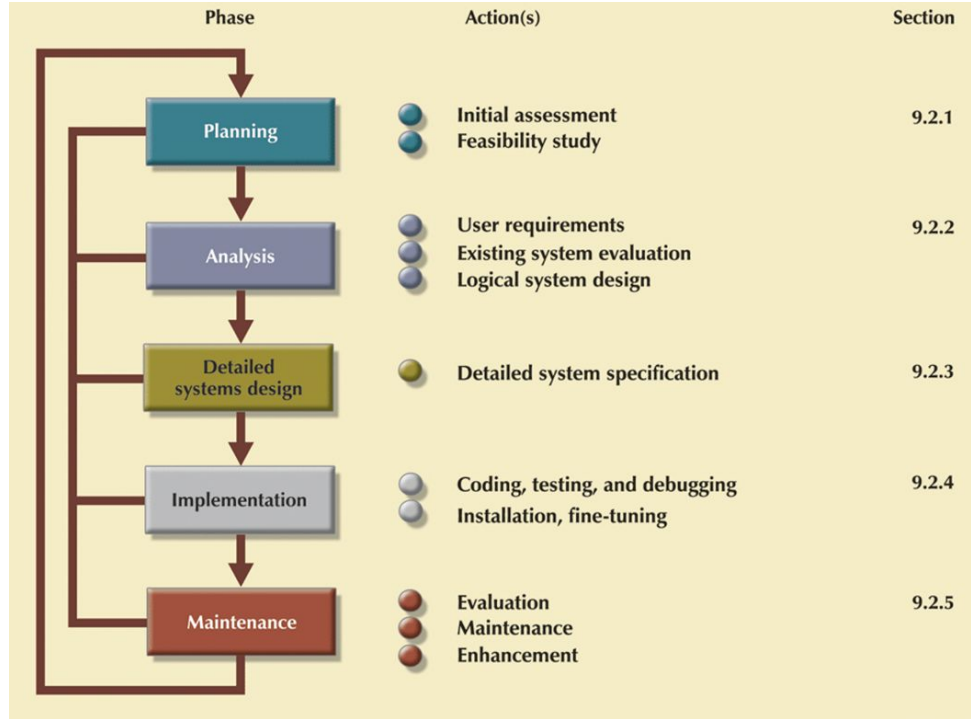
Performance depends on three factors:

- Database design and implementation
- Application design and implementation
- Administrative procedures

Database development is the process of database design and implementation

- Implementation phase includes: Creating database storage structure, Loading data into the database, Providing for data management

The Systems Development Life Cycle (SDLC)



Traces history (life cycle) of information system. As shown, the database design and application development mapped out and evaluated.

Divided into following five phases:

- Planning
- Analysis
- Detailed systems design
- Implementation
- Maintenance

Iterative rather than sequential process.

Planning

- General overview of company and objectives
- Assessment of flow-and-extent requirements
 - Should the existing system be continued?
 - Should the existing system be modified?
 - Should the existing system be replaced?
- Study and evaluate alternative solutions
 - Technical aspects of hardware and software requirements
 - System cost
 - Operational cost

Analysis

- Problems defined during planning phase examined in greater detail during analysis
- Thorough audit of user requirements
- Existing hardware and software systems are studied
- Goal:
 - Better understanding of:
 - System's functional areas
 - Actual and potential problems
 - Opportunities

Detailed Systems Design

- Designer completes design of system's processes
- Includes all necessary technical specifications
- Steps laid out for conversion from old to new system
- Training principles and methodologies are also planned
 - Submitted for management approval

Implementation

- Hardware, DBMS software, and application programs are installed
 - Database design implemented
- Cycle of coding, testing, and debugging continues until database ready for delivery
- Database created and system customized
 - Creation of tables and views
 - User authorizations

Maintenance

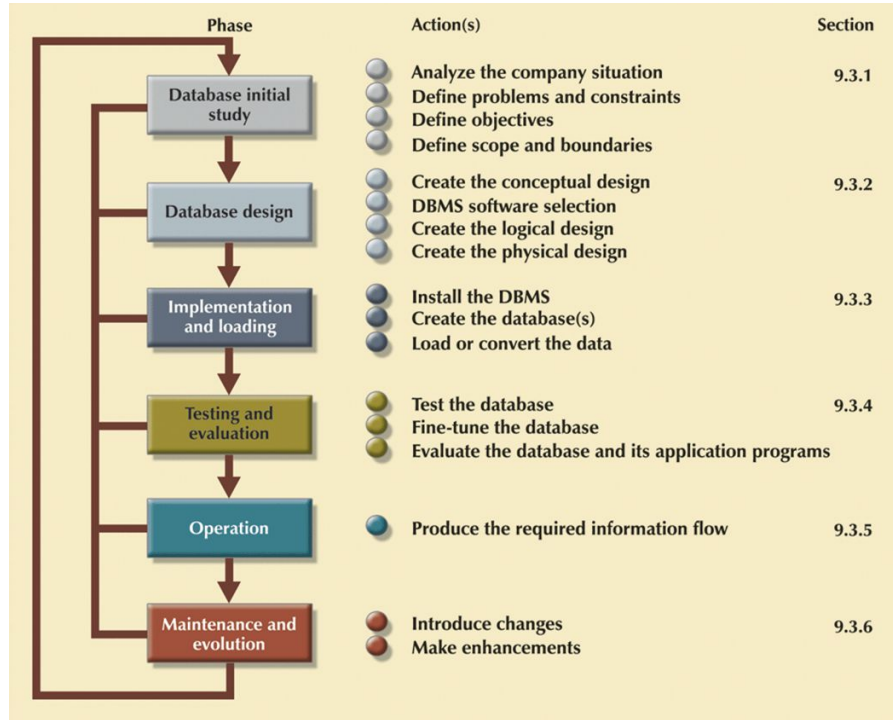
Three types of maintenance activity

- Corrective Maintenance
- Adaptive Maintenance
- Perfective Maintenance

Computer-aided systems engineering (CASE)

- Produce better systems within reasonable amount of time and at reasonable cost
- CASE-produced applications are structured, documented, standardized

The Database Life Cycle (DBLC)



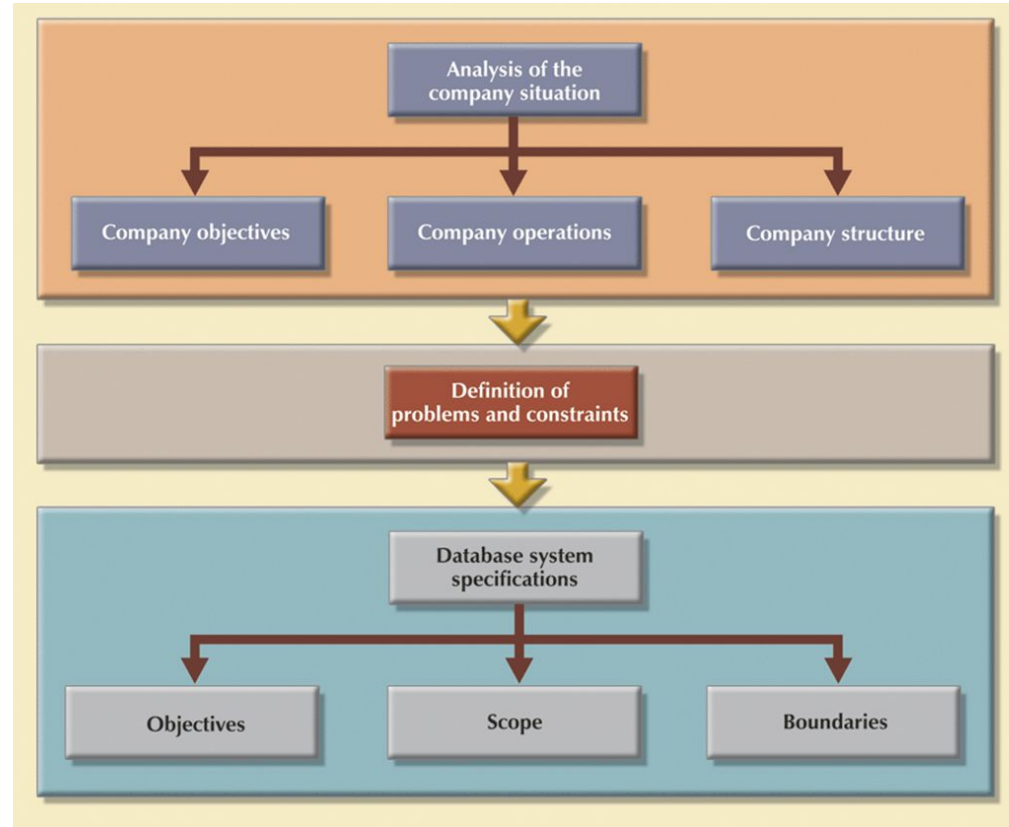
Six Phases

1. Database initial study
2. Database design
3. Implementation and loading
4. Testing and evaluation
5. Operation
6. Maintenance and evolution

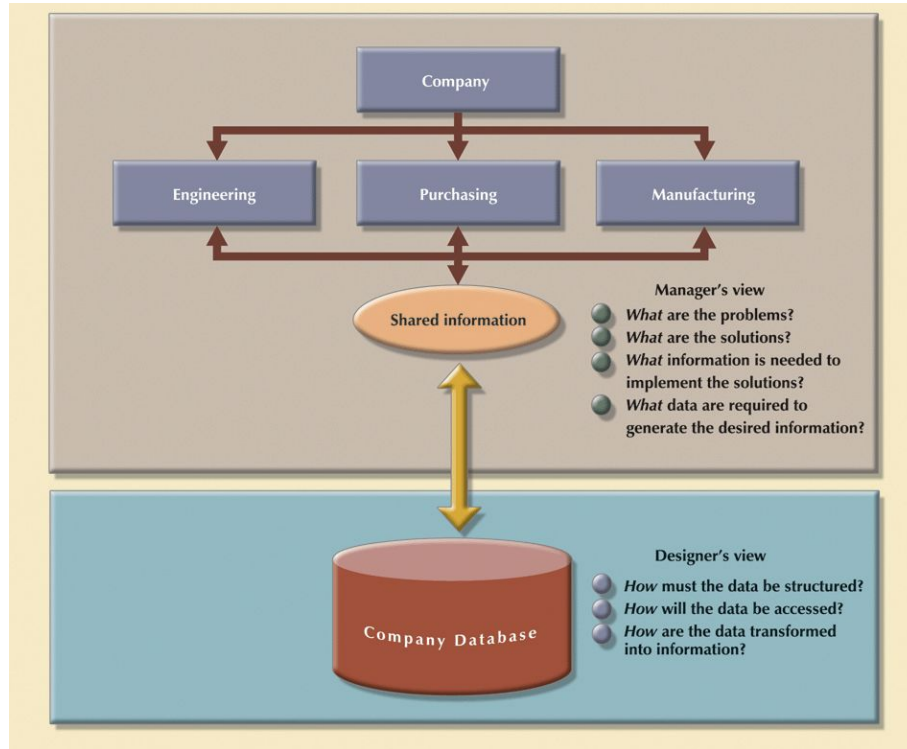
The Database Initial Study

Overall purpose:

- Analyze company situation
- Define problems and constraints
- Define objectives
- Define scope and boundaries
- Interactive and iterative processes required to complete first phase of DBLC successfully



Database Design



Database Design includes:

- Necessary to concentrate on data
- Characteristics required to build database model

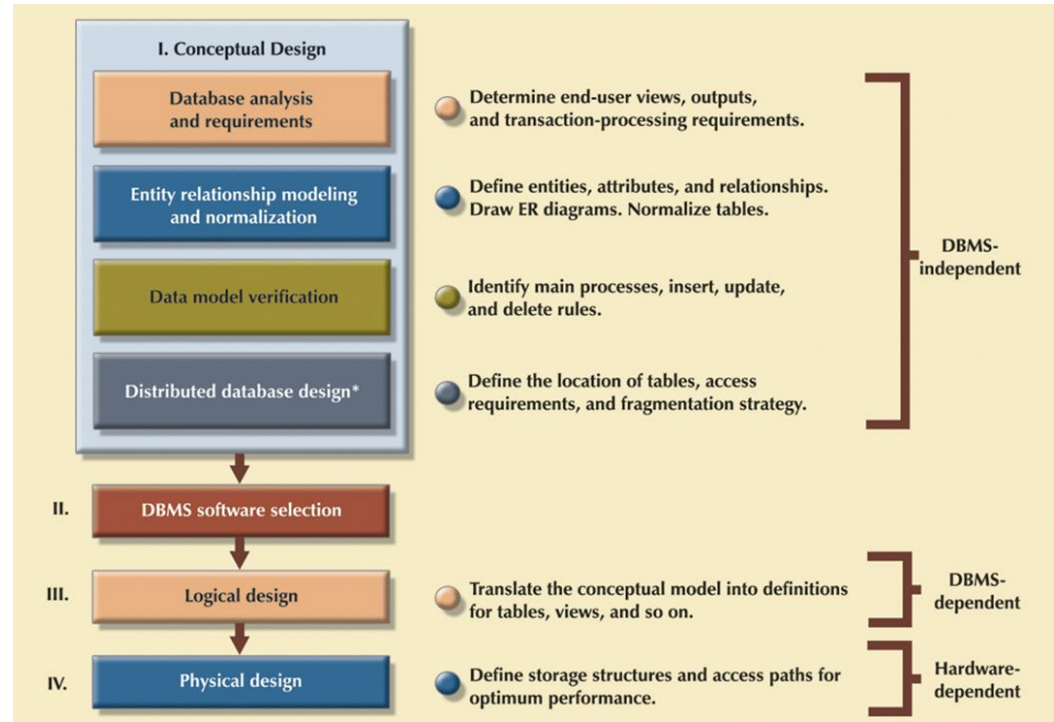
There are two views of data within the system:

- **Business view**
 - Data as information source
- **Designer's view**
 - Data structure, access, and activities required to transform data into information

Database Design Phase 1: Conceptual Design

Data modeling creates an abstract database structure; to represent real-world objects.

- Embodies clear understanding of business and its functional areas
- Ensure that all data needed are in model, and that all data in model are needed
- As shown in the figures, it requires four steps



01

Database Analysis and Requirements

- Discover data element characteristics
 - Obtains characteristics from different sources
- Take into account business rules
 - Derived from description of operations

02

Entity relationship modeling and normalization

- Designer enforces standards in design documentation
 - Use of diagrams and symbols, documentation writing style, layout, other conventions

03

Data model verification

- Verified against proposed system processes
- Revision of original design
- Define design's major components as modules (IS components for specific functions).

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Distributed Database Design

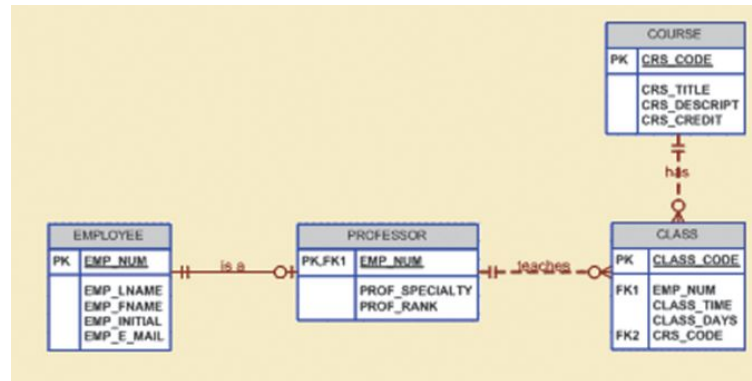
- Portions of database may reside in different physical locations; Processes accessing the database vary from one location to another
- Designer must also develop data distribution and allocation strategies

Database Design Phase 2: DBMS Software Selection

- Critical to information system's smooth operation
- Common factors affecting purchasing decisions:
 - Cost
 - DBMS features and tools
 - Underlying model
 - Portability
 - DBMS hardware requirements

Database Design Phase 3: Logical Design

- Translate conceptual design into internal model
- Logical design is software-dependent
- Requires all objects be mapped to specific constructs used by selected database software
 - Definition of attribute domains, design of required tables, access restriction formats
 - Tables must correspond to entities in conceptual design
- Translates software-independent conceptual model into software-dependent model



Simple conceptual model sample.

Database Design Phase 4: Physical Design

- Process of selecting data storage and data access characteristics of database
- Storage characteristics are function of:
 - Device types supported by hardware
 - Type of data access methods supported by system
 - DBMS
- More complex when data are distributed

Implementation and Loading

- New database implementation requires special storage-related constructs for end-user tables
- Performance
 - Difficult to evaluate
 - Not all DBMSs have tools embedded
 - No standard measurement
- Security:
 - Physical and password security
 - Access rights and audit trails
 - Data encryption and diskless workstations

Implementation and Loading (cont.)

- Backup and Recovery
 - Full backup
 - Differential backup
 - Transaction log backup
- Integrity
 - Enforced via proper use of primary, foreign key rules
- Company standards
 - Standards may be partially defined

Testing and Evaluation

- Occurs in parallel with applications programming
- Database tools used to prototype applications
- If implementation fails to meet some of system's evaluation criteria:
 - Fine-tune specific system and DBMS configuration parameters
 - Modify physical or logical design
 - Upgrade software and/or hardware platform

Operation

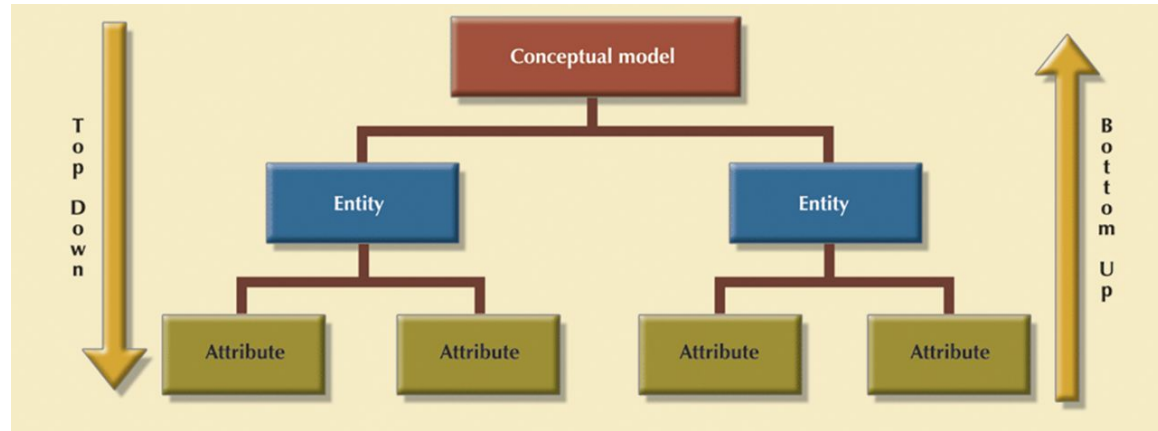
- Once database has passed evaluation stage, it is considered operational
- Beginning of operational phase starts process of system evolution
- Problems not foreseen during testing surface
- Solutions may include:
 - Load-balancing software to distribute transactions among multiple computers
 - Increasing available cache

Maintenance and Evaluation

Required periodic maintenance:

- Preventive maintenance (backup)
- Corrective maintenance (recovery)
- Adaptive maintenance
- Assignment of access permissions and their maintenance for new and old users
- Generation of database access statistics
- Periodic security audits
- Periodic system-usage summaries

Database Design Strategies

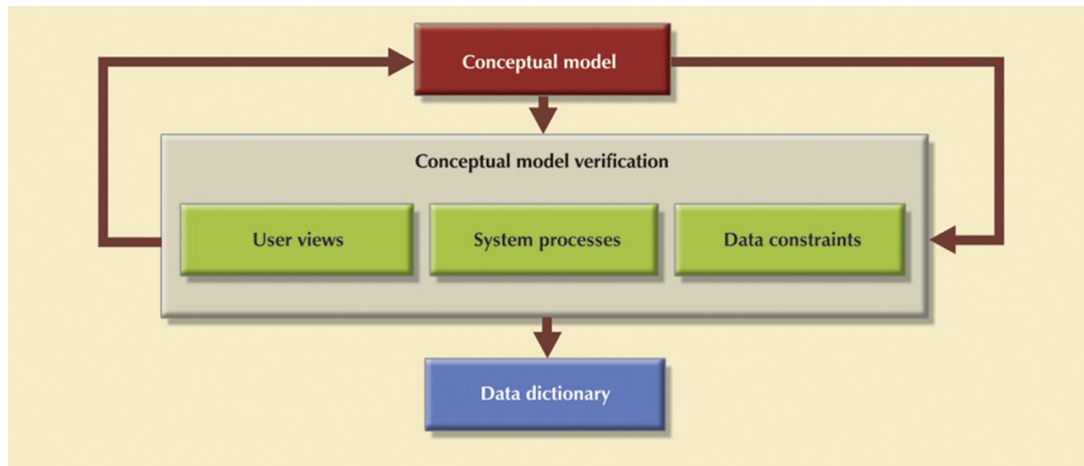


Top-down Design

- Identifies data sets
- Defines data elements for each of those sets
 - Definition of different entity types
 - Definition of each entity's attributes

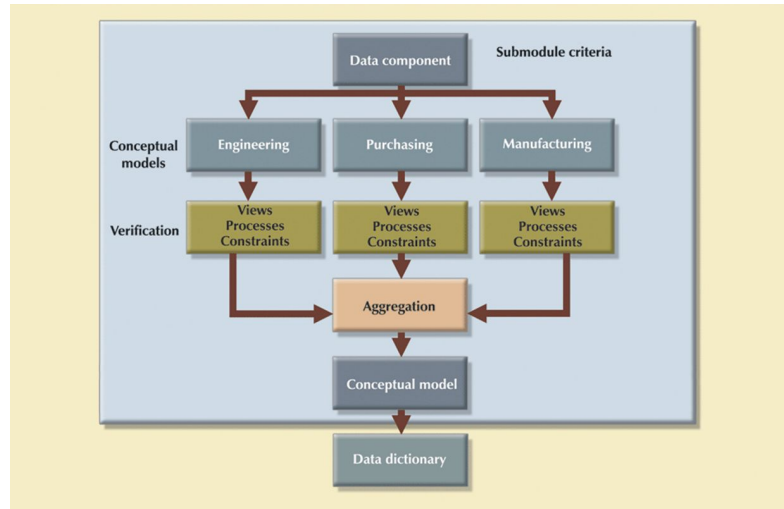
Bottom-up Design

- Identifies data elements (items)
- Groups them together in data sets



● Centralized design

- When data component is composed of small number of objects and procedures
- Typical of small systems

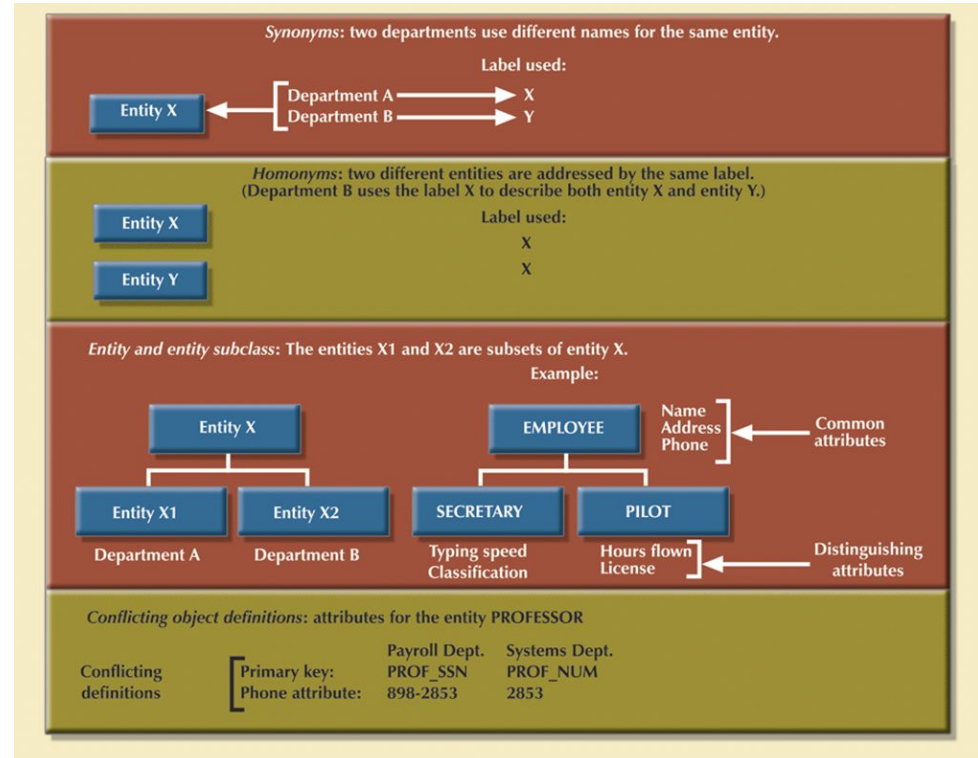


● Decentralized design

- Data component has large number of entities
- Complex relations on which complex operations performed
- Problem spread across several operational sites

Centralized vs Decentralized Design

- All modules integrated into one model
- Aggregation problems to be addressed:
 - Synonyms and homonyms
 - Entity and entity subtypes
 - Conflicting object definitions



Summary

- Information system facilitates transformation of data into information
 - Manages both data and information
- SDLC traces history (life cycle) of an application within the information system
- DBLC describes history of database within the information system
- Database design and implementation process moves through series of well-defined stages
- Conceptual design subject to several variations:
 - Top-down vs. bottom-up
 - Centralized vs. decentralized

Thank you for listening!
